

AI Career Coach - Project Documentation

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Frontend Architecture:

- app/(main)/dashboard/: DashboardPage and dashboard-view component
- app/(main)/resume/: ResumePage with resume-builder and entry-form components
- app/(main)/ai-cover-letter/: CoverLetterPage, NewCoverLetterPage, CoverLetterPreviewPage with cover-letter components
- app/(main)/roadmap/: RoadmapPage with roadmap components and error-boundary
- app/(main)/interview/: InterviewPrepPage, QuizPage with quiz components and charts
- app/(main)/onboarding/: OnboardingPage and onboarding-form
- components/: UI components (header, hero, button, card, input, label), theme and auth providers

Backend Architecture:

- Actions Folder:
- cover-letter.js: generateCoverLetter, getCoverLetters, getCoverLetter, deleteCoverLetter
- dashboard.js: generateAllInsights, getIndustryInsights
- interview.js: generateQuiz, saveQuizResult, getAssessments
- resume.js: saveResume, getResume, improveWithAI
- roadmap.js: generateRoadmap, getRoadmaps, updateRoadmapProgress
- user.js: updateUser, getUserOnboardingStatus

Database Design (Prisma Models):

- User: id, email, password, name, imageUrl, industry, bio, experience, skills[], related to assessments, coverLetter, resume, insights, roadmaps
- Assessment: id, userId, quizScore, questions[], category, improvementTip, createdAt
- Resume: id, userId, content, atsScore, feedback, createdAt
- CoverLetter: id, userId, content, jobDescription, companyName, jobTitle, status, createdAt
- IndustryInsight: id, industry, salaryRanges[], growthRate, demandLevel, topSkills[], marketOutlook, keyTrends[], recommendedSkills[], lastUpdated, nextUpdate

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- Roadmap: id, userId, industry, role, description, skills[], steps (JSON), timeframe, progress, status, createdAt

API Documentation:

- Auth: POST /api/auth/register, GET/POST /api/auth/[...nextauth]
- User: GET/PUT /api/user/profile
- Resume: POST /api/resume/save, GET /api/resume
- Cover Letter: POST /api/cover-letter, GET/DELETE /api/cover-letter/:id
- Interview: POST /api/interview/quiz, POST /api/interview/result, GET /api/interview/assessments
- Roadmap: POST /api/roadmap, GET /api/roadmap, PUT /api/roadmap/:id/progress
- Other: /api/debug, /api/health, /api/test-env, /api/inngest

Key Features:

- Auth with NextAuth.js and hashed passwords
- AI-generated cover letters and resumes
- Step-based career roadmaps and AI-driven quizzes
- Market insights per industry
- Guided onboarding and real-time UI feedback
- All data scoped to authenticated users

Implementation Details:

- API calls using fetch with JSON
- Future support for file uploads via API
- Code example: `fetch('/api/resume/save', POST, headers, body)`

Deployment Setup:

- Hosted on Vercel with vercel.json
- .env includes DATABASE_URL, NEXTAUTH_SECRET, NEXTAUTH_URL, GEMINI_API_KEY
- Scripts: setup-db.js

Performance Optimizations:

- Frontend: Lazy loading, code splitting, optimized assets
- Backend: Connection pooling, indexed queries, async operations

Security Measures:

- Bcrypt password hashing
- Session-based route protection
- Env vars for secrets, Zod validation
- Rate limiting and CORS (optional)

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Future Enhancements:

- Resume and certificate uploads
- Google/LinkedIn login
- Real-time features (WebSockets)
- Admin panel and analytics
- Mobile version, multiple AI models

Conclusion:

AI Career Coach is a secure, scalable, and AI-powered career platform. Its modular architecture and strong foundation make it easy to scale and extend for various job-seeking needs.